

Human Review: Dales Question (II) for $c_0 \oplus \ell_\infty$

Original automatic proof: `solution_packet.pdf`

Checked date: 15 June 2026

Status: Verified.

Verdict: The proof appears essentially correct. It gives a positive answer to the stated subquestion for

$$E = c_0 \oplus \ell_\infty,$$

namely that every finitely-generated maximal left ideal of $\mathcal{B}(E)$ is fixed.

Human Comments

The proof has a clear structure. The main new step is the following implication: if L is a non-fixed maximal left ideal of $\mathcal{B}(c_0 \oplus \ell_\infty)$ and L contains a bounded-below operator, then

$$\mathcal{K}_1 \subseteq L,$$

where

$$\mathcal{K}_1 = \left\{ \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathcal{B}(c_0 \oplus \ell_\infty) : A \in \mathcal{K}(c_0) \right\}.$$

This is then combined with Proposition 7.8 of the source paper, which states that $\mathcal{K}_1$ is not contained in any finitely-generated maximal left ideal of $\mathcal{B}(c_0 \oplus \ell_\infty)$.

The block argument looks sound. Since L is non-fixed, the strong dichotomy in the source paper gives

$$\mathcal{E}(E) \subseteq L,$$

where $\mathcal{E}(E)$ is the ideal of inessential operators. The required block inclusions are justified as follows. Compact operators on c_0 are inessential. Every operator $\ell_\infty \rightarrow c_0$ is strictly singular, because c_0 contains no copy of ℓ_∞ , and hence is inessential. The converse off-diagonal direction $c_0 \rightarrow \ell_\infty$ is also inessential by the standard symmetry property for inessential operators, exactly as used in the proof of Theorem 7.7 of the source paper via Gonzalez. Finally, weakly compact operators on ℓ_∞ are inessential.

Now let

$$R = \begin{pmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{pmatrix} \in L$$

be bounded below. Restricting R to the ℓ_∞ -summand gives the column operator

$$y \mapsto (R_{12}y, R_{22}y), \quad \ell_\infty \rightarrow c_0 \oplus \ell_\infty,$$

which is bounded below and hence upper semi-Fredholm. Since $R_{12} : \ell_\infty \rightarrow c_0$ is strictly singular, stability of upper semi-Fredholm operators under strictly singular perturbations implies that R_{22} is upper semi-Fredholm.

Choosing a finite-rank projection Q onto $\ker R_{22}$, the restriction of R_{22} to $\ker Q$ is an isomorphism onto its closed range. Since ℓ_∞ is injective, the inverse on this range extends to an operator $S \in \mathcal{B}(\ell_\infty)$, giving

$$SR_{22} = I_{\ell_\infty} - Q.$$

Multiplying R on the left by the lower-right block operator with entry S then produces

$$\begin{pmatrix} 0 & 0 \\ SR_{21} & I_{\ell_\infty} - Q \end{pmatrix} \in L.$$

The lower-left block SR_{21} is inessential, and Q is finite-rank. Since these corresponding block operators already belong to L , it follows that

$$\begin{pmatrix} 0 & 0 \\ 0 & I_{\ell_\infty} \end{pmatrix} \in L.$$

This forces all lower-right blocks into L , and together with the inessential upper-left compact and off-diagonal blocks gives $\mathcal{K}_1 \subseteq L$.

It remains to explain why a non-fixed finitely-generated maximal left ideal contains such a bounded-below operator. If

$$L = \mathcal{L}_\Gamma, \quad \Gamma = \{T_1, \dots, T_n\},$$

then the dichotomy theorem gives $\mathcal{F}(E) \subseteq L$. Corollary 4.7 of the source paper then implies that

$$\Psi_\Gamma : E \rightarrow E^n, \quad \Psi_\Gamma x = (T_1 x, \dots, T_n x),$$

is bounded below. Since $E^n \cong E$, choose an isomorphism $A : E^n \rightarrow E$. Then

$$R = A\Psi_\Gamma$$

is bounded below. Moreover $R \in L$, because writing A coordinatewise as

$$A(y_1, \dots, y_n) = A_1 y_1 + \dots + A_n y_n$$

gives

$$A\Psi_\Gamma = A_1 T_1 + \dots + A_n T_n \in \mathcal{L}_\Gamma = L.$$

Thus a non-fixed finitely-generated maximal left ideal would contain a bounded-below operator, hence would contain $\mathcal{K}_1$, contradicting Proposition 7.8. The conclusion follows.

Review Outcome

Accepted as an essentially correct proof of the $c_0 \oplus \ell_\infty$ subcase. The result should be viewed as a corollary-style completion of this specific open subquestion using the machinery already developed in the source paper, rather than as a solution to the broader remaining $C(K)$ cases.